Student Academic Performance Prediction Using Machine Learning

Machine Learning - Final Project Report

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Dataset Source: Kaggle - Student Performance Data by Devansodariya

1. Summary

This project develops a machine learning solution to predict student academic performance using a public Student Performance dataset from Kaggle. The main objective is to classify students into Pass or Fail categories based on academic, demographic, social, and behavioral factors. The original final grade column G3 was transformed into a binary target where students with G3 greater than or equal to 10 were labeled as Pass, and the others as Fail. The data was cleaned, categorical features were encoded, numerical features were scaled, and the dataset was split into training and testing sets. Four classification algorithms were trained and compared: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. The best performance was achieved by Logistic Regression and SVM, both reaching an F1-score of 0.90 and accuracy of 0.8734. The results show that previous grades are the strongest indicators of final performance.

2. Introduction

2.1 Motivation

Student academic performance is an important topic in educational data analysis because early prediction can help teachers and institutions identify students who may need additional support. In many schools and universities, students' success is influenced by many factors, such as study time, previous grades, attendance, family background, support systems, and personal habits. Machine learning can be used to discover patterns in these factors and provide useful predictions that support better decision-making.

The research question addressed in this project is: Can machine learning models predict whether a student will pass or fail based on available student-related features? This question is important because a prediction model can help educators focus intervention efforts on students at risk of poor performance.

2.2 Background

The project belongs to supervised machine learning because the dataset includes labeled examples: each student has a known final grade. The task is classification rather than regression because the numerical grade G3 was converted into two classes: Pass and Fail. This matches the project proposal and allows the use of classification evaluation measures such as accuracy, precision, recall, F1-score, and confusion matrix.

Several models studied in the course were selected for comparison. Logistic Regression is a simple and interpretable linear classifier. Decision Tree creates rule-based decisions that are easy to understand. Random Forest is an ensemble method that combines multiple trees to improve stability and generalization. Support Vector Machine attempts to find a decision boundary that separates classes with a good margin. Comparing these models helps us understand which approach works better for student performance prediction.

3. Dataset

3.1 Source

The dataset used in this project is Student Performance Data obtained from Kaggle. The file used in the implementation is student_data.csv. It contains student information and grades, and it is suitable for studying how different academic and social factors relate to final student performance.

3.2 Description

The dataset contains 395 student records and 33 original columns. After creating the target column Pass_Fail, the working dataset contains 34 columns. The original target grade is G3, which represents the final grade of the student on a scale from 0 to 20. For this project, G3 was converted into a binary classification target: Pass if G3 >= 10 and Fail if G3 < 10. This produced 265 Pass cases and 130 Fail cases.

The features include numerical variables such as age, studytime, failures, absences, G1, and G2, as well as categorical variables such as school, sex, address, family size, parent jobs, school support, family support, internet access, and desire for higher education.

Figure 1. Target distribution after converting G3 into Pass/Fail classes.

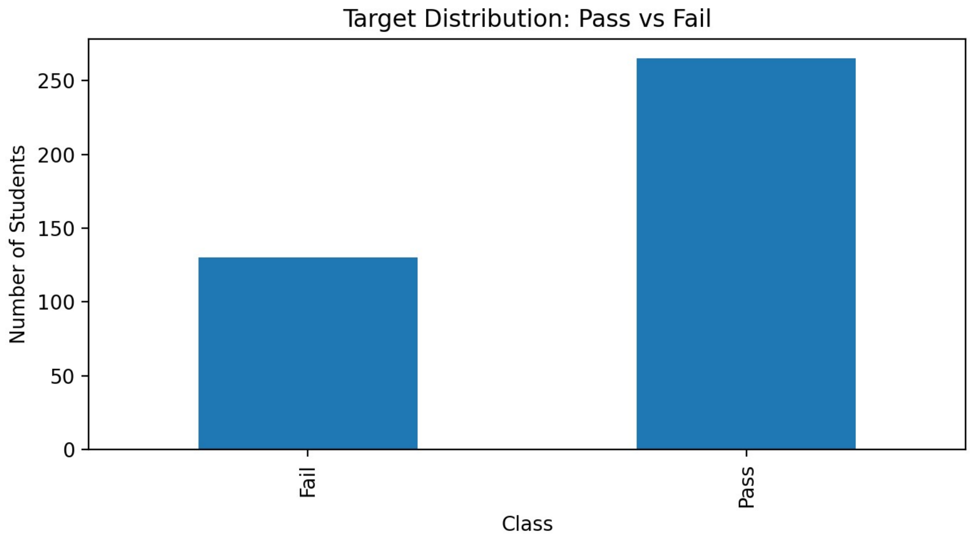


Table 1. Dataset overview.

Item	Value
Number of rows	395
Original columns	33
Target variable	Pass_Fail created from G3
Pass definition	G3 >= 10
Number of Pass students	265
Number of Fail students	130
Missing values	0

3.3 Preprocessing

The preprocessing steps were designed to make the data ready for machine learning algorithms. First, the target variable was created from G3. Then G3 was removed from the feature set to avoid using the answer directly. Categorical variables were transformed using One-Hot Encoding, while numerical variables were standardized using StandardScaler. The dataset was split into 80% training data and 20% testing data using stratified sampling to preserve the Pass/Fail class distribution. No missing values were found in the dataset.

The features G1 and G2 were kept in the main experiment because they represent previous academic performance and are valid predictors in an educational prediction context. However, an additional experiment was also conducted without G1 and G2 to show how the results change when previous grades are removed.

4. Methodology

4.1 Approach

The project follows an end-to-end machine learning workflow: load the data, explore the dataset, create the target variable, preprocess features, split the data, train several models, tune selected hyperparameters, evaluate the models, and interpret the results. The implementation was done in Python using pandas, NumPy, matplotlib, and scikit-learn.

Four models were selected because they represent different classification approaches: Logistic Regression as a baseline linear model, Decision Tree as an interpretable rule-based model, Random Forest as an ensemble model, and SVM as a margin-based classifier. These models are appropriate because the project is a binary classification problem with a relatively small tabular dataset.

Table 2. Models used in the project.

Model	Purpose	Main tuning parameters
Logistic Regression	Linear baseline classifier	C
Decision Tree	Rule-based interpretable classifier	max_depth, min_samples_leaf
Random Forest	Ensemble classifier using multiple trees	n_estimators, max_depth, min_samples_leaf
SVM	Margin-based classifier	C, kernel

4.2 Evaluation Criteria

The models were evaluated using accuracy, precision, recall, F1-score, and confusion matrix. Accuracy measures the overall percentage of correct predictions. Precision measures how many predicted Pass cases were actually Pass. Recall measures how many actual Pass students were correctly detected. F1-score combines precision and recall into one balanced metric. The confusion matrix shows correct and incorrect predictions for both Fail and Pass classes.

F1-score was used for hyperparameter tuning because the dataset is not perfectly balanced, with more Pass cases than Fail cases. Using F1-score helps evaluate both precision and recall instead of relying only on accuracy.

5. Results

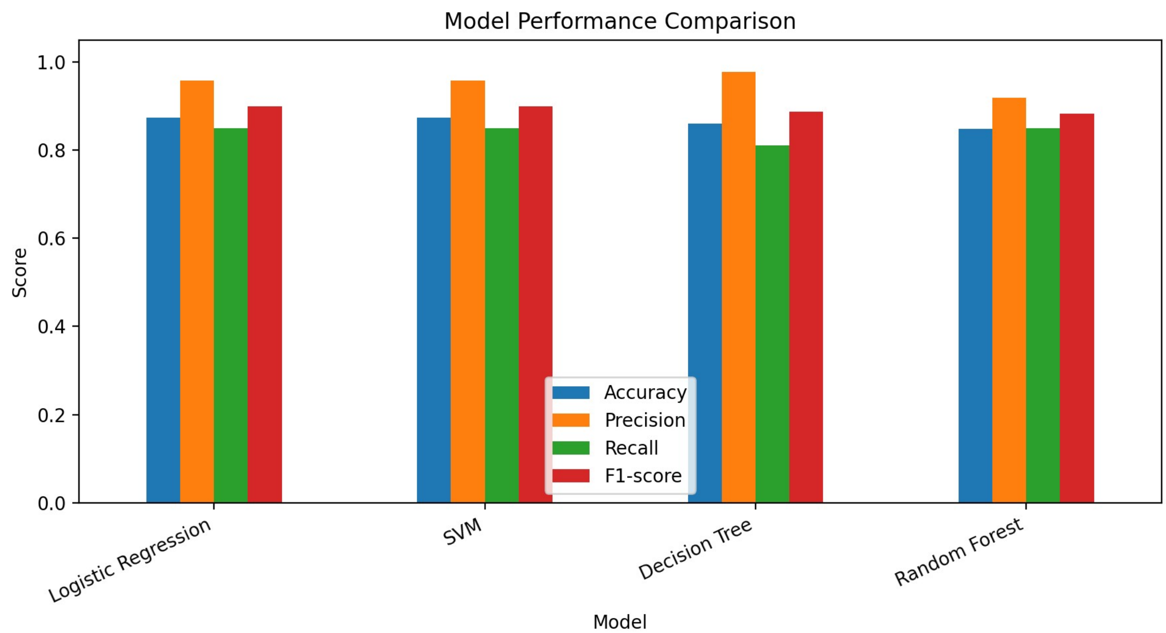
5.1 Model Comparison

Table 3. Model performance on the test set.

Model	Best Params	CV F1	Accuracy	Precision	Recall	F1-score
Logistic Regression	C=1	0.9218	0.8734	0.9574	0.8491	0.9
SVM	C=10, linear	0.921	0.8734	0.9574	0.8491	0.9

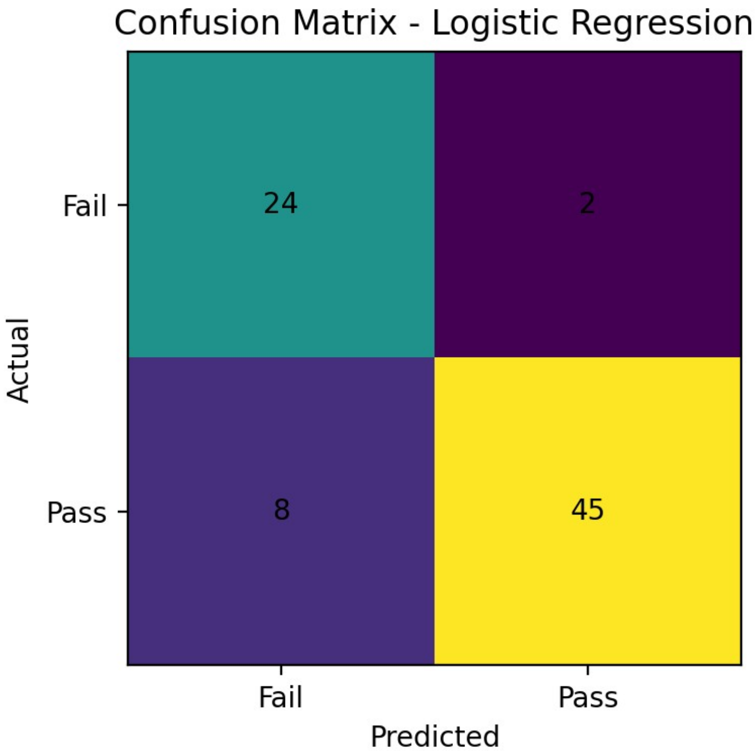
Model	Best Params	CV F1	Accuracy	Precision	Recall	F1-score
Decision Tree	depth=None, leaf=1	0.9393	0.8608	0.9773	0.8113	0.8866
Random Forest	n=50, depth=None, leaf=5	0.9337	0.8481	0.9184	0.8491	0.8824

Figure 2. Comparison of classification metrics for the four models.



The results show that Logistic Regression and SVM achieved the best test performance, both with an accuracy of 0.8734 and F1-score of 0.9000. Decision Tree achieved an F1-score of 0.8866, while Random Forest achieved an F1-score of 0.8824. These results indicate that linear decision boundaries work well for this dataset, especially when previous grades G1 and G2 are included.

Figure 3. Confusion matrix for the best model, Logistic Regression.



For the best model, the confusion matrix shows 24 correctly predicted Fail students, 45 correctly predicted Pass students, 2 Fail students incorrectly predicted as Pass, and 8 Pass students incorrectly predicted as Fail. This means the model performs well overall, but still misses some students who actually pass.

5.2 Feature Findings

Figure 4. Top numeric correlations with the final grade G3.

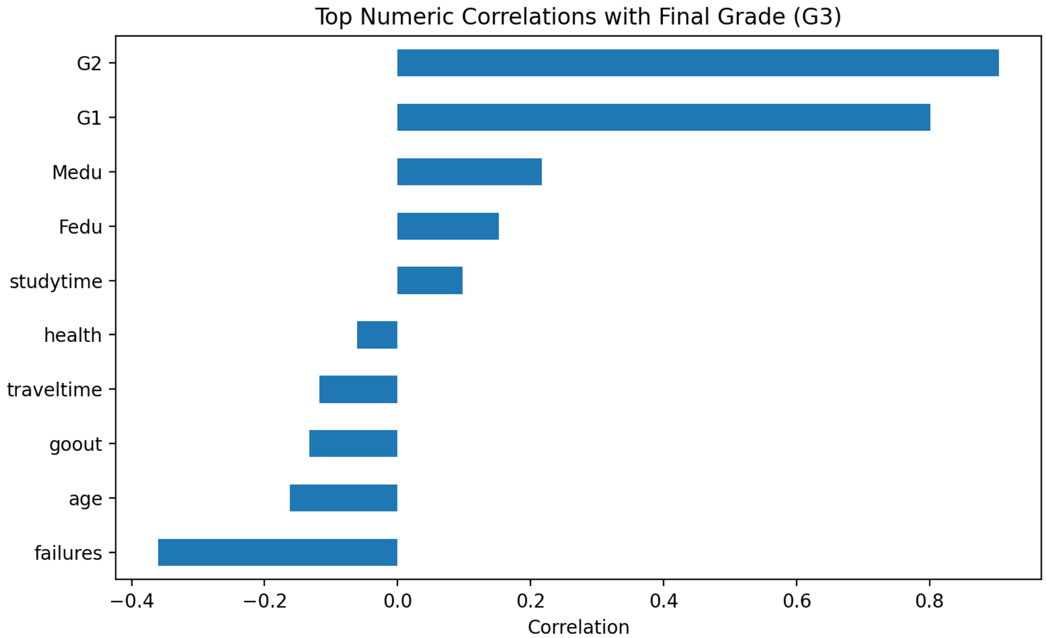


Figure 5. Permutation feature importance for the best model.

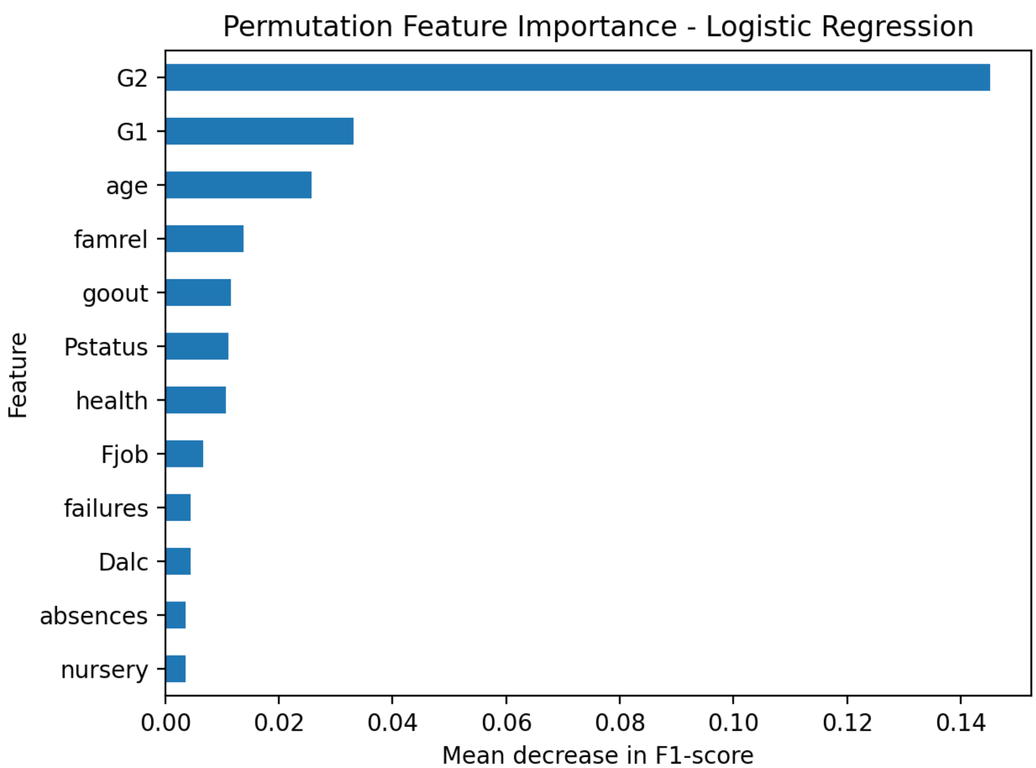


Table 4. Most important features according to permutation importance.

Feature	Importance
G2	0.1452

Feature	Importance
G1	0.0331
age	0.0257
famrel	0.0138
goout	0.0116
Pstatus	0.0111
health	0.0107
Fjob	0.0067
failures	0.0045
Dalc	0.0044
absences	0.0036
nursery	0.0036

The feature analysis confirms that previous grades G1 and G2 are the most influential predictors. This is expected because they measure earlier academic performance and are closely related to the final grade G3. Other factors, such as failures, absences, studytime, and school or family support, may also contribute to performance, but their effect is weaker compared with previous grades.

5.3 Additional Experiment Without G1 and G2

Table 5. Model performance when G1 and G2 are removed.

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.6709	0.68	0.9623	0.7969
Logistic Regression	0.6582	0.7167	0.8113	0.7611
SVM	0.6582	0.7167	0.8113	0.7611
Decision Tree	0.6456	0.6984	0.8302	0.7586

When G1 and G2 were removed, model performance decreased clearly. The best model in this scenario was Random Forest with an F1-score of 0.7969 and accuracy of 0.6709. This result shows that previous academic performance is the strongest source of predictive information. Without previous grades, the models still learn useful patterns, but the prediction task becomes more difficult.

6. Discussion

The main finding is that student final performance can be predicted with good accuracy when previous grades and student-related features are available. Logistic Regression and SVM performed best in the main experiment, suggesting that the relationship between the strongest features and the target is mostly linearly separable. Decision Tree and Random Forest also performed well, but they did not improve over the simpler linear models on the test set.

The high importance of G1 and G2 is both a strength and a limitation. It is a strength because previous grades are realistic and useful predictors for educational intervention. However, it is also a limitation because using previous grades can make the task easier and may hide the effects of non-grade factors such as study time, absences, and family support. For this reason, the additional experiment without G1 and G2 was important. It showed that the models become less accurate when previous grades are removed, meaning that early academic results are the best warning signal for final success or failure.

Another limitation is the small size of the dataset. It contains only 395 student records, so the results may not generalize to all schools or education systems. A larger dataset with more recent records and more diverse students would improve reliability. Also, the project uses binary classification, which simplifies performance into Pass and Fail. A future improvement could predict multiple levels such as Low, Medium, and High performance, or use regression to predict the exact final grade.

7. Conclusion

This project successfully built and evaluated machine learning models for predicting student academic performance. The dataset was prepared by cleaning the data, encoding categorical variables, scaling numerical variables, and converting the final grade G3 into a Pass/Fail classification target. Logistic Regression, Decision Tree, Random Forest, and SVM were trained and compared using accuracy, precision, recall, F1-score, and confusion matrix.

The best results were achieved by Logistic Regression and SVM, both reaching an F1-score of 0.90 and accuracy of 0.8734 on the test set. The analysis showed that previous grades G1 and G2 are the strongest predictors of final performance. When these grades were removed, performance dropped, confirming their importance. Overall, the project demonstrates how machine learning can support educational analysis and help identify students who may need academic support. Future work could use a larger dataset, more advanced models, and multiclass or regression prediction to provide deeper insights.

8. Team Contribution

The project was completed collaboratively by the group members. Both members participated in understanding the project requirements, selecting the dataset, preparing the data, building the machine learning models, evaluating the results, and writing the final report. The team discussed the work through meetings and communication, and the final decisions were made jointly. This ensured that both members understood the dataset, preprocessing steps, model training process, evaluation results, and the main conclusions.

9. Presentation Outline

The presentation should be 5-10 minutes and can be organized as follows:

- Slide 1: Project title, names, and project objective.

- Slide 2: Problem motivation and why predicting student performance is useful.
- Slide 3: Dataset source, number of rows and columns, and target variable.
- Slide 4: Preprocessing steps: Pass/Fail target, encoding, scaling, and train/test split.
- Slide 5: Models used: Logistic Regression, Decision Tree, Random Forest, and SVM.
- Slide 6: Evaluation metrics: accuracy, precision, recall, F1-score, and confusion matrix.
- Slide 7: Results table and performance comparison chart.
- Slide 8: Feature importance and the role of G1/G2.
- Slide 9: Conclusion, limitations, and future work.

10. References

1. Devansodariya. Student Performance Data. Kaggle dataset. https://www.kaggle.com/datasets/devansodariya/student-performance-data
2. Scikit-learn Documentation. Machine Learning in Python. https://scikit-learn.org/
3. Pandas Documentation. Python Data Analysis Library. https://pandas.pydata.org/
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